

PEDRO LOURES · UNITY DEVELOPER

# Pedro Loures

I turn game designs into systems that ship.

Unity developer since 2012: gameplay systems, VR interaction, editor and Blender tools, networking and data pipelines, for studios, schools and clients like Fiat and Vale. Some of it is still in use today. Most recently I was Unity lead on Anchored Memories, a full-body haptic VR experience at UT Austin.

Unity · C#

Gameplay systems

Tools · Python

Networking & data

VR / Haptics

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## SELECTED PROJECTS



HAPTIC VR · UNITY · 2026

### Anchored Memories

Haptic VR journey on climate migration · Unity lead · UT Austin, 2026

**Role** Unity Lead (D2Studios, volunteer) · **Year** 2026 · **Engine** Unity · **Team** 12+ (TXI, D2Studios, UT Austin advisors) · **For** Texas Immersive Institute, UT Austin

- Unity lead for D2Studios on a haptic VR experience built around the bHaptics TactSuit X40
- Contributed assets from my own library, bought over the years, to speed up production

#### LEADERSHIP

- Moved the project to Unity: the team had started planning in Unreal, and switched to my main engine when I joined so we could reach a working deliverable for the July 2026 showcase



MANAGEMENT SIM · UNITY · 2021

### Dream Arcade Simulator

Arcade management sim · Centennial College capstone, team of 5

**Role** Gameplay Programmer · **Year** 2021 · **Engine** Unity · **Team** 5 · **For** Centennial College capstone

- Day/night cycle with skybox, exposure and light blending
- The whole night phase: shop with tabs, grid placement, move, rotate and sell
- Audio manager with fade-in/out jobs, and a data-driven cheat code system using UnityEvents
- Fixed an AI overheating issue caused by objects that weren't being disposed



VR · 4 MINIGAMES · SOLE PROGRAMMER

## VR Carnival

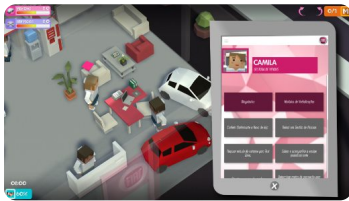
Four seated VR minigames in under a month · sole programmer

**Role** Sole Programmer, Game Design · **Engine** Unity · SteamVR · **Team** 3 · **For** Centennial College VR course

- MiniGameManager singleton base class with overridable lifecycle hooks, so only one minigame runs at a time
- Dissolve shader effect for materializing the bow and hammer in the player's hand
- Extended SteamVR's interaction samples (grabbing, bow, buttons) into four distinct games

### LEADERSHIP

- Scoped a short-handed team to four core games in under a month, then iterated on polish



TRAINING SIM · CLIENT: FIAT

## Fiat Quality Manager Training

Data-driven training simulation for Fiat dealership Quality Managers

**Engine** Unity · **For** Fiat · D2R Studios

- Content pipeline: Fiat's Excel scenario sheets → JSON → NoSQL database, downloaded by Unity at runtime
- 76 real-world scenarios at launch, with a web admin so the client can add more without a new build



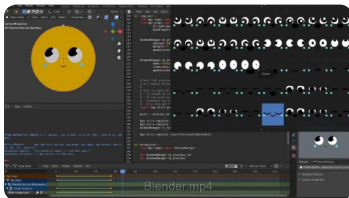
MULTI-SCREEN APP · CLIENT: VALE · STILL IN USE

## Vale · Porto de Tubarão

Three-screen presentation system controlled from a tablet · 2012 → 2019

**Role** Desktop developer (2012) · Rebuild developer (2019) · **Year** 2012 / 2015 / 2019 · **Engine** Unity · **For** Vale · Zor Design · D2R Studios

- Custom sprite system written before Unity had native 2D
- Content pipeline through the Resources folder: new images only need the right file names, with no code or level changes
- Rebuilt the tablet app from scratch and replaced OnGUI with real networking (UNet) in one month for the 2019 version
- Still in use today, more than a decade after the first version



ANIMATION TOOL · PYTHON · STILL IN USE

## Blender Plugin

Python add-on that speeds up facial-expression keyframing for animators

**Engine** Blender · Python

- Lists every facial texture for a character's face parts, read from a folder naming convention, so animators pick expressions from one panel
- Saves around 3 clicks per keyframe when swapping faces, plus a one-click keyframe button
- Built as a side project alongside my regular work, fitted into the studio's existing animation pipeline, and still in use today



HORROR · GAME JAM · TEAM LEAD

## Coma · College Game Jam

Horror game jam on the theme “Another World” · led a team of newer developers

**Role** Team Lead · Programmer · **Year** 2021 · **Engine** Unity · **For** College midterm game jam

- Procedural maze generator (adapted from Kodeco’s maze tutorial) whose output was saved and then decorated by hand
- Reusable trigger system built on UnityEvents, with cooldowns and one-shot options

### LEADERSHIP

- The coordinator placed me with newer students so I could lead; I set up a prefab workflow that let the artists do level design themselves



EDUCATIONAL · LUA · UNITY

## Mitologia da Programação

Educational game where players solve Greek-myth challenges by writing Lua

**Engine** Unity · embedded Lua · **Team** 2

- Embedded a Lua interpreter in Unity, so players’ code compiles and runs inside the game
- Checks each solution against the expected output, with custom hint messages or the compiler’s own errors



PUZZLE PLATFORMER · SBGAMES AWARD · 2013

## Loneliness

Shadow-puzzle platformer · SBGames 2013: 3rd in Popular Vote + Best Short Paper

**Year** 2013 · **For** PUC capstone

- Replay system for the shadows, rebuilt from frame-data recording to an input-state machine to survive frame drops

## MORE WORK

Animal Defense, Fiat Personalização, Custom Defense, Procedural Dungeon Generator, Chaveirinho, Digital Wonderland, Recifuturo. Details, videos and code at [pedroloures.com/projects](https://pedroloures.com/projects).

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